

Identification-Robust Inference for Complier Cumulative-Incidence Effects under Competing Risks

Naresh Garg

19 August 2026

Abstract

Instrumental-variable analyses of competing-risk outcomes commonly estimate a cause-specific complier cumulative-incidence effect by dividing an intention-to-treat contrast in Aalen-Johansen cumulative incidences by an intention-to-treat contrast in treatment uptake. This ratio is regular only when compliance is bounded away from zero. We propose a confidence set obtained by inverting the studentized, undivided moment for a candidate complier effect. The procedure uses the joint influence representation of the two Aalen-Johansen estimators and the two treatment proportions, never divides by the estimated first stage, and remains valid along triangular-array sequences in which the compliance contrast is of order $n^{-1/2}$. The causal estimand, identification result, and score-inversion principle are established; the contribution is their finite-grid Aalen-Johansen specialization, including the exact tied-time influence representation. A reproducible simulation compares the score set with the conventional ratio-Wald interval. The score set maintains near-nominal coverage and becomes uninformative when the data contain little information, whereas the ratio estimator and its interval can be numerically unstable. The method is deliberately limited to a randomized binary instrument without baseline covariate adjustment.

Keywords: Aalen-Johansen estimator; competing risks; instrumental variable; local average treatment effect; weak identification.

1 Introduction and precise contribution

Competing events make a binary endpoint at a fixed horizon scientifically meaningful but statistically coarsened by right censoring. Examples include relapse before death and stroke before noncardiovascular death. When treatment is confounded but a randomized encouragement is available, the cause-specific cumulative-incidence effect among compliers is identified by a Wald-type ratio (Richardson et al., 2017). The numerator is the instrument contrast in the cause-specific cumulative incidence, and the denominator is the instrument contrast in treatment uptake.

The ratio representation creates an inferential problem that is separate from identification. If compliance is small, linearizing the ratio requires division by a quantity close to zero. A symmetric Wald interval can then be unstable, extremely long, or misleadingly regular-looking. Weak-instrument-robust test inversion is classical for linear IV models and has recently been extended to nonparametric LATE with high-dimensional covariates (Ma, 2026) and analyzed directly for influence-function score sets (Smucler, Lanni, and Masip, 2025). Those results do not themselves supply the joint right-censored competing-risk score used here.

Research question. For one binary randomized instrument, one binary treatment, one event cause, and one fixed horizon, can inference for the complier cumulative-incidence risk difference remain valid when the compliance contrast is $O(n^{-1/2})$?

Contribution. We combine the finite-grid Aalen-Johansen influence representation with an undivided LATE moment and prove uniform pointwise-in-effect coverage under local-to-zero compliance. This is an outcome-specific specialization of established Fieller/Anderson-Rubin inversion, not a new causal estimand, identification theorem, or generic score-inversion principle. Its scientific value is that loss of identification is reported as a wide, possibly full feasible confidence set rather than hidden by a ratio approximation.

2 Related literature and novelty comparison

Richardson et al. (2017) identified and estimated the exact cause-specific complier cumulative-incidence contrast with a binary IV and right-censored competing risks. Kjaersgaard and Parner (2016) used pseudo-observations and generalized method of moments for survival, restricted mean survival, and cumulative incidence. Lee, Kennedy, and Mitra (2023) developed influence-function-based, doubly robust estimators for complier survival effects under covariate- and outcome-dependent censoring, but did not study cause-specific cumulative incidence or weak identification. Model-based competing-risk IV approaches include Zheng et al. (2017) and Beyhum et al. (2023).

Weak-identification-robust ratio inference substantially predates this paper. Fieller (1954) obtained confidence sets for ratios by inverting a studentized undivided contrast, and Anderson and Rubin (1949) developed the corresponding weak-instrument-robust principle for linear IV models. Ma (2026) established identification-robust inference for nonparametric LATE with high-dimensional covariates, while Smucler, Lanni, and Masip (2025) characterized the possible forms of the generic score-inverted LATE confidence set. Accordingly, neither score inversion nor its quadratic confidence-set topology is new. The present contribution is the finite-grid Aalen–Johansen specialization, including the exact tied-time influence representation and its joint covariance with treatment uptake.

Conference literature addresses several adjacent but distinct IV problems. Kuang et al. (AISTATS 2020) synthesize candidate instruments; Hartford et al. (ICML 2021) study estimation when some candidate instruments violate exclusion; Liu et al. (CLear 2022) use predicted compliance to construct data-driven exclusion criteria; and Ailer, Hartford, and Kilbertus (ICML 2023) study underspecified high-dimensional instrument selection. None of these papers supplies weak-identification-robust inference for a right-censored cause-specific complier CIF. This literature search does not prove the absence of an unlocated predecessor. Table 1 summarizes the closest methods and the remaining gap.

Table 1: Closest journal and conference literature and the remaining inferential gap.

Work	Assumptions	Estimand/data	Method	Limitation relative to this paper
Richardson et al. (2017)	Binary valid IV; monotonicity; independent censoring	Cause-specific complier CIF; right-censored competing risks	Aalen-Johansen Wald ratio	Conventional ratio inference; no local-to-zero first-stage result
Kjaersgaard and Parner (2016)	Structural moment and censoring conditions	Survival, RMST, CIF	Pseudo-observation GMM	Not a nonparametric complier-CIF weak-IV set
Lee et al. (2023)	Conditional valid IV; censoring models	Complier survival probability	Doubly robust IF estimators	No competing-risk CIF or weak-IV theory
Fieller (1954); Anderson–Rubin (1949)	Joint regular limit for an undivided contrast	Ratio or linear-IV parameter	Test inversion	Not an Aalen–Johansen competing-risk construction
Ma (2026)	Nonparametric LATE; high-dimensional covariates	Fully observed scalar outcome	Orthogonal identification-robust test	No censoring or competing risks
Smucler et al. (2025)	Generic nonparametric LATE	Fully observed outcome	Inverted IF score	No Aalen-Johansen competing-risk score
Kuang et al. (AISTATS 2020)	Majority/structure restrictions on candidate IVs	General downstream causal effect	Instrument synthesis	Instrument construction, not weak-IV CIF inference

Table 1 continued

Work	Assumptions	Estimand/data	Method	Limitation relative to this paper
Hartford et al. (ICML 2021)	Modal validity among candidate IVs	General/CATE	Ensemble over IV estimators	Invalid-IV problem, not right censoring or weak identification
This paper	Randomized binary IV; monotonicity; exclusion; independent censoring	Fixed-horizon cause-specific complier CIF	Joint AJ/uptake score inversion	No covariates; one horizon; relies on valid IV and censoring assumptions

3 Setup, estimand, and assumptions

For independent subjects $i = 1, \dots, n$, observe $O_i = (Z_i, A_i, \tilde{T}_i, \Delta_i)$. The randomized instrument Z and received treatment A are binary. Let T be the event time, $J \in \{1, \dots, K\}$ its cause, C the censoring time, $\tilde{T} = \min(T, C)$, and $\Delta = \mathbb{I}(T \leq C)$, using an event-wins-ties convention. Fix cause j and horizon τ .

Let A^z denote treatment under instrument z , and define $Y_j^a(\tau) = \mathbb{I}(T^a \leq \tau, J^a = j)$. The target is

$$\psi_j(\tau) = E\{Y_j^1(\tau) - Y_j^0(\tau) \mid A^1 > A^0\}.$$

We assume: (A1) consistency, no interference, and well-defined treatment versions; (A2) Z is randomized with $P(Z = z) = \pi_z \in (\epsilon, 1 - \epsilon)$; (A3) exclusion, so Z affects (T, J) only through A ; (A4) monotonicity $A^1 \geq A^0$ almost surely; (A5) $\kappa = P(A = 1 \mid Z = 1) - P(A = 1 \mid Z = 0) > 0$, although κ may approach zero with n ; (A6) $C \perp (T, J) \mid Z$ through τ , with the conditional censoring survival bounded away from zero. For the theoretical result, follow-up through τ is recorded on a fixed finite grid $\mathcal{T}_\tau = \{t_1, \dots, t_L\}$, where L does not increase with n . Define

$$Y_i(k) = \mathbb{I}(\tilde{T}_i \geq t_k), \quad N_{i\ell}(k) = \mathbb{I}(\tilde{T}_i = t_k, \Delta_i = \ell), \quad N_{i\cdot}(k) = \sum_{\ell=1}^K N_{i\ell}(k).$$

For $z \in \{0, 1\}$, let

$$H_z(k) = E\{Y_i(k) \mid Z_i = z\}, \quad \alpha_{z\ell}(k) = \frac{E\{N_{i\ell}(k) \mid Z_i = z\}}{H_z(k)}, \quad \alpha_{z\cdot}(k) = \sum_{\ell=1}^K \alpha_{z\ell}(k).$$

We additionally assume: (A7) for fixed constants $h_0, a_0 > 0$,

$$\min_{z,k} H_z(k) \geq h_0, \quad \max_{z,k} \alpha_{z\cdot}(k) \leq 1 - a_0;$$

and (A8), at the true effect, the variance of the score influence function defined below is bounded away from zero. Assumptions A6–A8 match the unadjusted finite-grid Aalen–Johansen estimator. Censoring dependent on treatment or baseline covariates requires a different, adjusted observed-data score and is outside the present theorem.

Define $F_{zj}(\tau) = P(T \leq \tau, J = j \mid Z = z)$, $p_z = P(A = 1 \mid Z = z)$, $N_j = F_{1j} - F_{0j}$, and $\kappa = p_1 - p_0$. Under A1–A5, the established LATE identity is

$$\psi_j(\tau) = N_j(\tau)/\kappa.$$

The CIF is a total effect on the probability of cause j by τ . It includes pathways through changes in competing-event incidence and is not a controlled direct effect.

4 Proposed methodology

Within each instrument arm, estimate $F_{zj}(\tau)$ by the Aalen-Johansen estimator

$$\hat{F}_{zj}(\tau) = \sum_{u \leq \tau} \hat{S}_z(u-) d\hat{\Lambda}_{zj}(u),$$

and estimate p_z by the arm-specific treatment proportion. Set $\hat{N}_j = \hat{F}_{1j} - \hat{F}_{0j}$ and $\hat{\kappa} = \hat{p}_1 - \hat{p}_0$.

Write

$$S_z(t_k-) = \prod_{r < k} \{1 - \alpha_{z\cdot}(r)\}, \quad F_{zj}(t_k) = \sum_{r \leq k} S_z(t_r-) \alpha_{zj}(r).$$

The conditional-arm influence function of the finite-grid Aalen-Johansen estimator is

$$\begin{aligned} \phi_{F,z}(O) &= \sum_{k=1}^L \frac{S_z(t_k-)}{H_z(k)} \{N_j(k) - Y(k) \alpha_{zj}(k)\} \\ &\quad - \sum_{k=1}^L \frac{F_{zj}(\tau) - F_{zj}(t_k)}{H_z(k) \{1 - \alpha_{z\cdot}(k)\}} \{N_{\cdot}(k) - Y(k) \alpha_{z\cdot}(k)\}. \end{aligned} \quad (1)$$

Here $F_{zj}(t_k)$ includes the cause- j increment at t_k , so the second term concerns increments strictly after t_k . The factor $\{1 - \alpha_{z\cdot}(k)\}^{-1}$ is the finite-jump correction required for tied discrete event times. The full-sample influence functions are

$$\Phi_{F,z}(O) = \frac{I(Z=z)}{\pi_z} \phi_{F,z}(O), \quad \Phi_{p,z}(O) = \frac{I(Z=z)}{\pi_z} (A - p_z),$$

with $\Phi_N = \Phi_{F,1} - \Phi_{F,0}$ and $\Phi_\kappa = \Phi_{p,1} - \Phi_{p,0}$.

Let $\hat{\Phi}_{N,i}$ and $\hat{\Phi}_{\kappa,i}$ be the joint plug-in influence values, retaining their covariance because the same subjects contribute treatment and event information. For candidate effect $b \in [-1, 1]$, define

$$\hat{g}_i(b) = \hat{\Phi}_{N,i} - b \hat{\Phi}_{\kappa,i}, \quad T_n(b) = \frac{n(\hat{N}_j - b\hat{\kappa})^2}{n^{-1} \sum_i \hat{g}_i(b)^2}.$$

The 1-alpha confidence set is

$$\mathcal{C}_{1-\alpha} = \{b \in [-1, 1] : T_n(b) \leq \chi_{1,1-\alpha}^2\}.$$

The set may be the full feasible range or, algebraically, disconnected. Software must therefore store the acceptance indicator over the grid or solve the quadratic inequality, rather than silently replacing a disconnected set by its convex hull.

5 Theoretical results

Proposition 1 (Identification; established). *Under A1–A5 and $\kappa > 0$, $\psi_j(\tau) = N_j(\tau)/\kappa$.*

Proof. Randomization and consistency express the instrument-arm contrast as $E[Y_j^{A^1}(\tau) - Y_j^{A^0}(\tau)]$. Exclusion and monotonicity make this contrast zero for always-takers and never-takers and equal to $E[(Y_j^1(\tau) - Y_j^0(\tau))I(A^1 > A^0)]$. The treatment contrast is $P(A^1 > A^0)$. Division gives the result. This is the standard LATE argument specialized to the cause- j indicator and is not new. $\square$

For a law P , define

$$h_b(P) = F_{1j}(\tau) - F_{0j}(\tau) - b(p_1 - p_0) = N_j(\tau) - b\kappa.$$

Let $\mathcal{P}$ denote the class of laws satisfying A1–A8 with the same constants ϵ, h_0, a_0 , and suppose

$$0 < \underline{\sigma}^2 \leq \text{var}_P\{g_{\psi(P),P}(O)\} \leq \bar{\sigma}^2 < \infty$$

uniformly over $P \in \mathcal{P}$. No uniform positive lower bound is imposed on $\kappa(P)$.

Theorem 1 (Finite-grid identification-robust inference). *For every possibly n -dependent sequence $P_n \in \mathcal{P}$ with $\kappa(P_n) > 0$, including sequences satisfying $\kappa(P_n) = c/\sqrt{n}$ or $\kappa(P_n) = o(n^{-1/2})$,*

$$\sqrt{n}\{\hat{h}_b - h_b(P_n)\} = \frac{1}{\sqrt{n}} \sum_{i=1}^n g_{b,P_n}(O_i) + o_{P_n}(1)$$

uniformly over $b \in [-1, 1]$. Moreover,

$$\sup_{b \in [-1, 1]} \left| \frac{1}{n} \sum_{i=1}^n \hat{g}_i(b)^2 - E_{P_n} \{g_{b,P_n}(O)^2\} \right| \xrightarrow{P_n} 0.$$

Consequently, for $\psi(P_n) = N_j(P_n)/\kappa(P_n)$,

$$\sup_{P \in \mathcal{P}} \left| \Pr_P \{T_n(\psi(P)) > \chi_{1,1-\alpha}^2\} - \alpha \right| \rightarrow 0,$$

and

$$\inf_{P \in \mathcal{P}} \Pr_P \{\psi(P) \in \mathcal{C}_{1-\alpha}\} \rightarrow 1 - \alpha.$$

Interpretation. The theorem gives coverage of the true effect uniformly over the stated distribution class and instrument strengths; it is not a simultaneous confidence band over multiple horizons or causes. It does not assert useful power when κ_n is small. An uninformative full set is correct behavior near nonidentification.

6 Simulation design

The executable Python program generates $X, U \sim N(0, 1)$, $Z \sim \text{Bernoulli}(0.5)$, and monotone treatment using one shared $V \sim \text{Uniform}(0, 1)$: $q_0 = \text{expit}(-1 + 0.45X + 0.55U)$, $q_1 = q_0 + \delta(1 - q_0)$, and $A^z = I(V \leq q_z)$. Figure 1 summarizes the simulated dependence structure; U and V are latent to the analysis. Two event causes follow ten-period multinomial hazards with linear predictors

$$\eta_{1k} = -3.35 + 0.10k - 0.45A + 0.35X + 0.55U,$$

$$\eta_{2k} = -3.55 + 0.07k + 0.25A - 0.25X + 0.45U.$$

The horizon is $\tau = 8$. Independent discrete censoring is calibrated to 0% or 30% by period 10, and events win censoring ties. Because $q_1 - q_0 = \delta(1 - q_0)$, the complier distribution of (X, U) does not depend on δ . Product Gauss–Hermite quadrature therefore gives one deterministic truth for every positive instrument strength:

$$\psi = \frac{E[(1 - q_0)\{F_1^1(8 | X, U) - F_1^0(8 | X, U)\}]}{E(1 - q_0)} = -0.0964917269, \quad E(1 - q_0) = 0.7114055309.$$

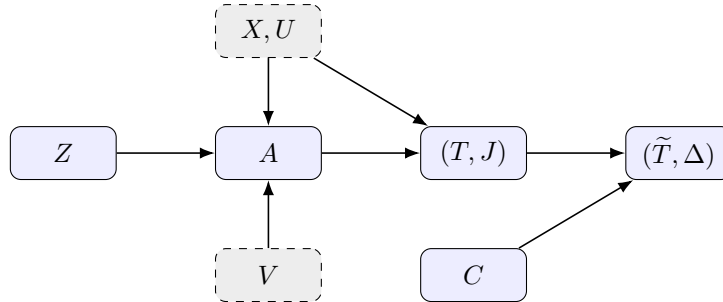


Figure 1: Simulation data-generating structure. Dashed nodes are latent to the unadjusted estimator. In the primary design, C is generated independently; the informative-censoring stress test adds dependence of C on U .

The value is stable beyond 12 decimal places from 40 through 80 nodes.

The fixed-strength experiment uses $n \in \{500, 1000\}$, $\delta \in \{0.05, 0.15, 0.35\}$, and 0% or 30% censoring. A direct local-to-zero experiment uses

$$n \in \{500, 1000, 2500, 5000\}, \quad \delta_n = \frac{c}{0.7114055309\sqrt{n}}, \quad c \in \{0.5, 1, 2, 4\},$$

so that $\sqrt{n}\kappa_n = c$ exactly in expectation. Four additional valid-design cells use 50% or 65% independent censoring. Eight paired cells compare the exact finite-jump influence function with a continuous-time/no-tie approximation. Finally, two informative-censoring and two defier cells are labelled assumption-violation diagnostics and are not evidence for the theorem. The 28 primary fixed-strength and local-to-zero cells use 5,000 replications; the 16 supplementary cells use 2,000, for 172,000 datasets in total. The master seed is 20260819.

Comparators use the same reduced-form and first-stage estimates: a ratio-Wald interval divides by $\hat{\kappa}$, whereas the score set inverts the undivided moment. The code solves the quadratic inequality exactly and records every connected component. Primary outcomes are coverage, exact union length, full/disconnected/empty-set frequency, and the median, 90th, 95th, and 99th percentiles of Wald length.

7 Simulation results

Table 2 reports the fixed-strength results. A nominal coverage estimate has Monte Carlo standard error about 0.003. Score coverage is generally near 0.95 (0.9446–0.9534), with ordinary finite-sample variation. Weak instruments make the set uninformative: for $\delta = 0.05$, its median length is the full feasible width 2 and the full-set frequency is 0.721–0.806. Wald coverage is then close to one only because its intervals have extreme tails, not because they are informative. Means and ratio RMSE are omitted because rare $\hat{\kappa} \approx 0$ replications make them non-reproducible heavy-tail summaries.

Figure 2 is the direct check of the theorem’s local-to-zero regime. Across all 16 cells, score coverage is 0.9452–0.9554 and is approximately invariant in n at fixed c . Information, correctly, is governed by local strength: the full-set frequency is 0.851–0.858 at $c = 0.5$, 0.755–0.760 at $c = 1$, 0.428–0.457 at $c = 2$, and 0.021–0.027 at $c = 4$.

Figure 3 shows why a mean Wald length is a poor diagnostic. At $n = 1000$ and 30% censoring, the 99th percentile is 4136 for $\delta = 0.05$ but only 0.69 for $\delta = 0.35$; the score set is bounded by the feasible width 2. Figure 4 additionally shows that disconnected sets occur in up to 1.94% of weak-IV replications and empty feasible sets in up to 0.08%. Replacing the set by its convex hull would therefore change actual simulation output.

The additional valid-design experiments tell the same story. At 50% and 65% independent censoring, score coverage is 0.9405–0.9455; median length increases to 1.261–1.684. In the four paired tie ablations, the finite-jump correction changes estimated score coverage by only 0.0010–0.0025 at the hazards used, but it is retained because it is the correct influence representation for the discrete design. The informative-censoring and defier experiments are diagnostics only: their numerical coverage does not restore the causal interpretation lost when A6 or monotonicity fails.

Table 2: Fixed-strength Monte Carlo results (5,000 replications per cell). Length columns report medians except “Wald p99.” Score length is the exact measure of the accepted union within $[-1, 1]$.

δ	n	Cens.	Mean $\hat{\kappa}$	Score set			Ratio-Wald		
				Cov.	Length	Full	Cov.	Length	p99
0.05	500	0%	0.036	0.953	2.000	0.803	0.999	5.16	10582.5
0.05	500	30%	0.036	0.948	2.000	0.806	0.999	5.48	4606.3
0.05	1000	0%	0.035	0.949	2.000	0.721	0.999	3.76	2850.1
0.05	1000	30%	0.036	0.946	2.000	0.734	0.999	3.92	4135.8
0.15	500	0%	0.107	0.945	1.513	0.303	0.992	1.55	35.0
0.15	500	30%	0.106	0.945	1.588	0.318	0.993	1.69	34.1
0.15	1000	0%	0.107	0.947	1.159	0.071	0.978	1.08	3.8
0.15	1000	30%	0.106	0.949	1.203	0.086	0.982	1.16	3.9
0.35	500	0%	0.249	0.953	0.676	0.001	0.965	0.64	1.1
0.35	500	30%	0.250	0.949	0.717	0.001	0.964	0.67	1.2
0.35	1000	0%	0.249	0.945	0.461	0.000	0.951	0.45	0.6
0.35	1000	30%	0.250	0.952	0.489	0.000	0.960	0.47	0.7

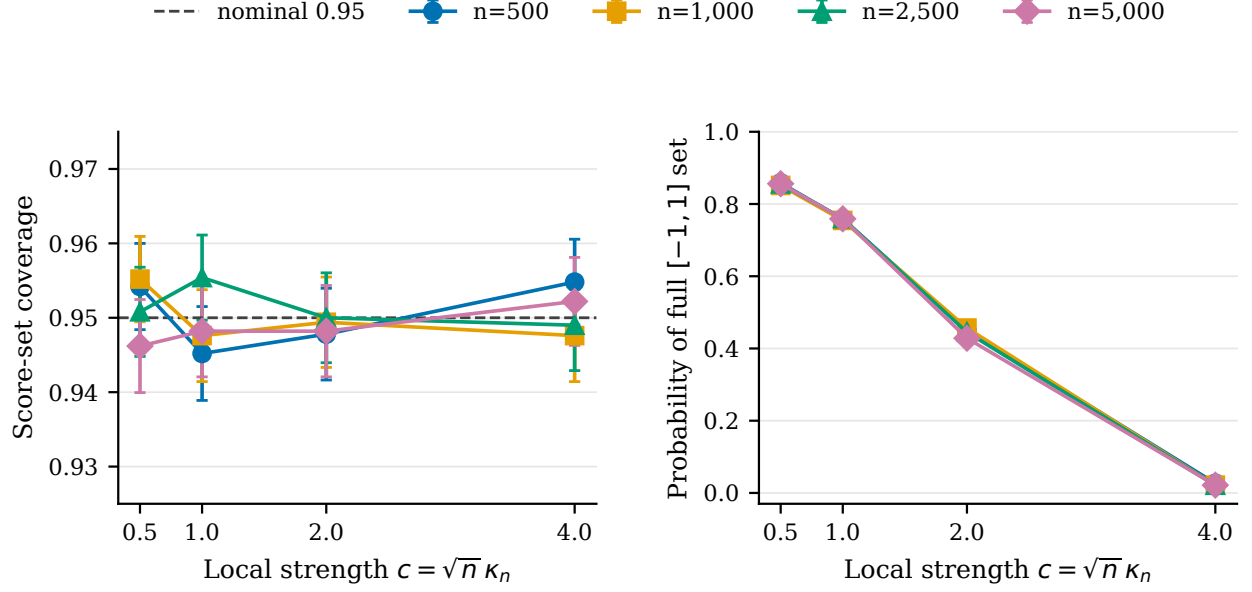


Figure 2: Local-to-zero experiment under 30% independent censoring. Left: score-set coverage with pointwise 95% Monte Carlo error bars. Right: probability of the full feasible set. Each point uses 5,000 replications and $\sqrt{n} \kappa_n = c$ in expectation.

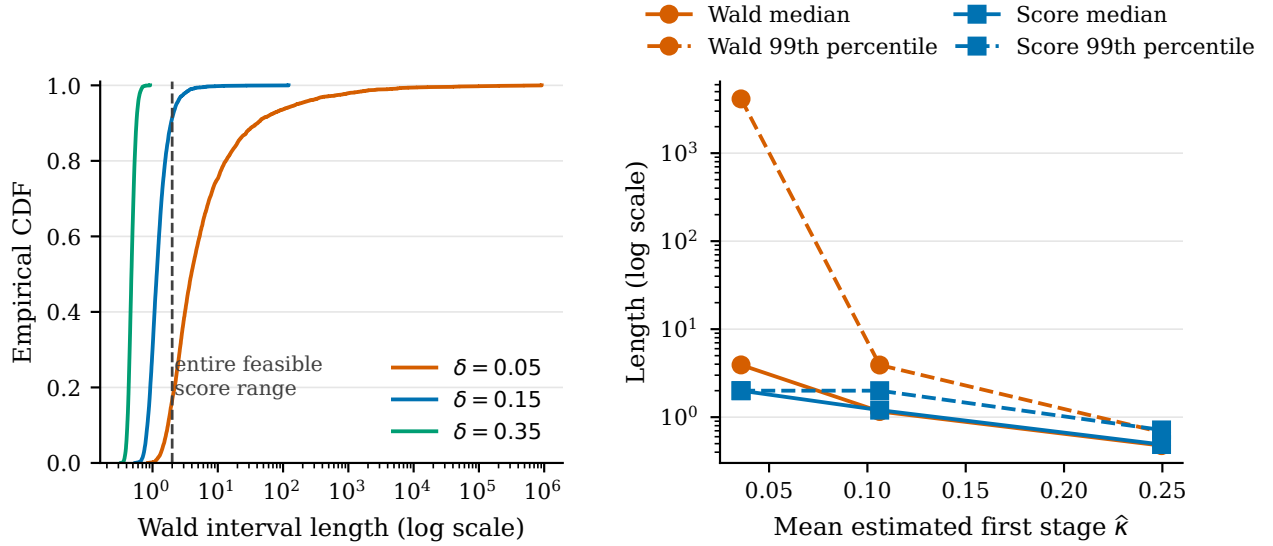


Figure 3: Near-zero-denominator pathology in the fixed-strength design ($n = 1000$, 30% censoring). Left: empirical distributions of Wald length. Right: median and 99th-percentile lengths for Wald intervals and score sets. Both axes reporting length use logarithmic scaling where indicated.

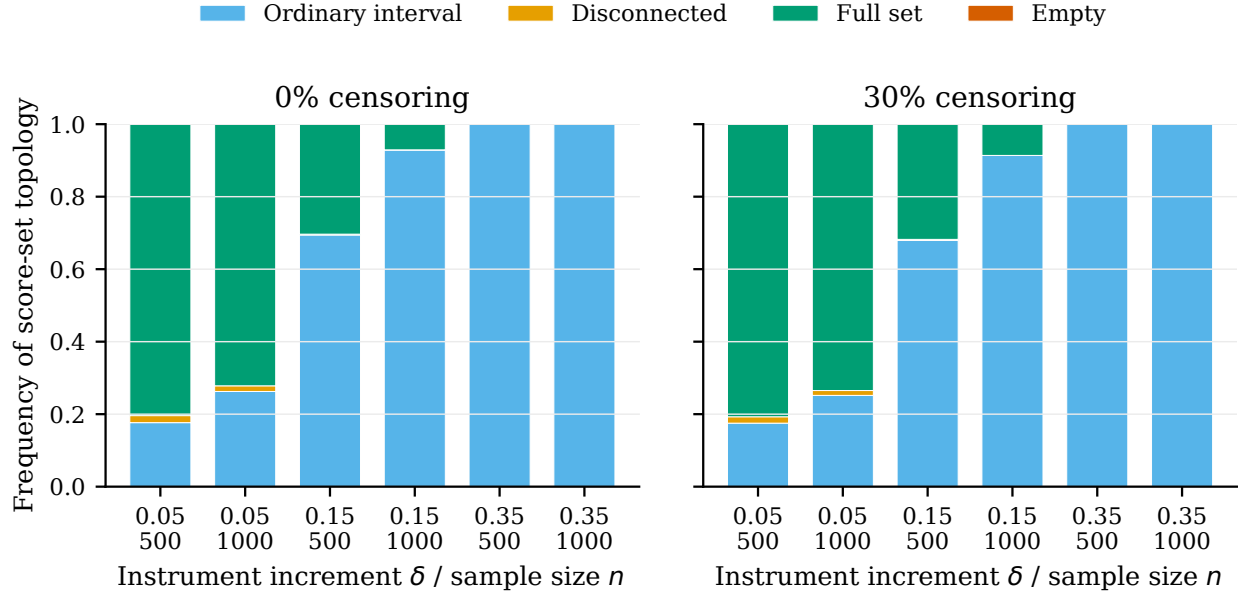


Figure 4: Exact topology of the score-inverted confidence set in the fixed-strength experiment. “Ordinary interval” excludes the full set; the small disconnected and empty categories are retained rather than collapsed to a convex hull.

8 Practical implications and limitations

Analysts should report the estimated compliance contrast, the score confidence set, whether the set is disconnected or full, and the competing-risk and censoring conventions. A narrow ratio-Wald interval should not be reported merely after screening on a first-stage threshold; such pretesting changes the inferential procedure. The method is appropriate for randomized encouragement studies or similarly defensible binary IVs with independent censoring within instrument arm.

The method should not be used without modification when censoring depends on treatment or covariates, instruments or providers create clustering, treatment is time varying, monotonicity is implausible, exclusion is violated, or the target is a direct effect unaffected by competing events. Covariate-adjusted and cross-fitted extensions are plausible but not proved here. Exact-zero compliance leaves the complier effect undefined; the theorem concerns sequences with positive κ_n .

9 Conclusion

For a fixed-horizon cause-specific complier cumulative-incidence effect, the inferentially consequential step is not another ratio estimator but avoidance of ratio linearization. Inverting the joint Aalen-Johansen/treatment-uptake score gives valid uncertainty under local-to-zero compliance, provided the usual competing-risk and IV assumptions hold. The price of honesty is an uninformative set when the design contains little causal information.

References

- Aalen, O. O., and Johansen, S. (1978). An empirical transition matrix for non-homogeneous Markov chains based on censored observations. *Scandinavian Journal of Statistics*, 5, 141-150.
- Ailer, E., Hartford, J., and Kilbertus, N. (2023). Sequential underspecified instrument selection for cause-effect estimation. *Proceedings of the 40th International Conference on Machine Learning*, PMLR 202. <https://proceedings.mlr.press/v202/ailier23a.html>
- Allignol, A., Schumacher, M., and Beyersmann, J. (2010). A note on variance estimation of the Aalen-Johansen

- estimator of the cumulative incidence function in competing risks, with a view towards left-truncated data. *Biometrical Journal*, 52, 126-137. <https://doi.org/10.1002/bimj.200900039>
- Anderson, T. W., and Rubin, H. (1949). Estimation of the parameters of a single equation in a complete system of stochastic equations. *Annals of Mathematical Statistics*, 20, 46-63. <https://doi.org/10.1214/aoms/1177730090>
- Beyersmann, J., Di Termini, S., and Pauly, M. (2013). Weak convergence of the wild bootstrap for the Aalen-Johansen estimator of the cumulative incidence function of a competing risk. *Scandinavian Journal of Statistics*, 40, 387-402. <https://doi.org/10.1111/j.1467-9469.2012.00817.x>
- Beyhum, J., Florens, J.-P., and Van Keilegom, I. (2023). A nonparametric instrumental approach to confounding in competing risks models. *Lifetime Data Analysis*, 29, 611-641. <https://doi.org/10.1007/s10985-023-09599-3>
- Fieller, E. C. (1954). Some problems in interval estimation. *Journal of the Royal Statistical Society: Series B (Methodological)*, 16(2), 175-185. <https://doi.org/10.1111/j.2517-6161.1954.tb00159.x>
- Hartford, J., Veitch, V., Sridhar, D., and Leyton-Brown, K. (2021). Valid causal inference with (some) invalid instruments. *Proceedings of the 38th International Conference on Machine Learning*, PMLR 139. <https://proceedings.mlr.press/v139/hartford21a.html>
- Kjaersgaard, M. I. S., and Parner, E. T. (2016). Instrumental variable method for time-to-event data using a pseudo-observation approach. *Biometrics*, 72, 463-472. <https://doi.org/10.1111/biom.12451>
- Kuang, Z., Sala, F., Sohoni, N., Wu, S., Cordova-Palomera, A., Dunnmon, J., Priest, J., and Re, C. (2020). Ivy: Instrumental variable synthesis for causal inference. *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics*, PMLR 108. <https://proceedings.mlr.press/v108/kuang20a.html>
- Lee, Y., Kennedy, E. H., and Mitra, N. (2023). Doubly robust nonparametric instrumental variable estimators for survival outcomes. *Biostatistics*, 24, 518-537. <https://doi.org/10.1093/biostatistics/kxab036>
- Liu, T., Lawlor, P., Ungar, L., and Kording, K. (2022). Data-driven exclusion criteria for instrumental variable studies. *Proceedings of the First Conference on Causal Learning and Reasoning*, PMLR 177. <https://proceedings.mlr.press/v177/liu22a.html>
- Ma, Y. (2026). Identification-robust inference for the LATE with high-dimensional covariates. *Journal of Econometrics*, 257, 106302. <https://doi.org/10.1016/j.jeconom.2026.106302>; [arXiv:2302.09756](https://arxiv.org/abs/2302.09756).
- Richardson, A., Hudgens, M. G., Fine, J. P., and Brookhart, M. A. (2017). Nonparametric binary instrumental variable analysis of competing risks data. *Biostatistics*, 18, 48-61. <https://doi.org/10.1093/biostatistics/kxw023>
- Smucler, E., Lanni, J., and Masip, D. (2025). A note on the properties of the confidence set for the local average treatment effect obtained by inverting the score test. [arXiv:2506.10449](https://arxiv.org/abs/2506.10449). <https://arxiv.org/abs/2506.10449>
- Zheng, C., Dai, R., Hari, P. N., and Zhang, M.-J. (2017). Instrumental variable with competing risk model. *Statistics in Medicine*, 36(8), 1240-1255. <https://doi.org/10.1002/sim.7205>

A Proof of the finite-grid theorem

Let $W(O)$ collect the finitely many bounded indicators

$$I(Z = z), \quad I(Z = z)A, \quad I(Z = z)Y(k), \quad I(Z = z)N_\ell(k),$$

for $z \in \{0, 1\}$, $k = 1, \dots, L$, and $\ell = 1, \dots, K$. All empirical arm probabilities, risk probabilities, hazards, treatment proportions, and Aalen-Johansen cumulative incidences are functions of

$$q_P = E_P\{W(O)\}, \quad \hat{q} = \mathbb{P}_n W.$$

On the set defined by A2 and A7, these functions are twice continuously differentiable, and their first and second derivatives are uniformly bounded. A uniform second-order Taylor expansion therefore gives

$$\sqrt{n}(\hat{\theta} - \theta_P) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \Phi_{\theta,P}(O_i) + R_{n,P}, \quad \sup_{P \in \mathcal{P}} \Pr_P(\|R_{n,P}\| > \eta) \longrightarrow 0$$

for every $\eta > 0$, where $\theta_P = (F_{1j}, F_{0j}, p_1, p_0)^\top$. Indeed, because W has fixed dimension and bounded coordinates, $\|\hat{q} - q_P\| = O_P(n^{-1/2})$ uniformly, while the Taylor remainder is $O_P\{\sqrt{n}\|\hat{q} - q_P\|^2\} = O_P(n^{-1/2})$.

The influence function of the arm-specific hazard is

$$\frac{I(Z = z)}{\pi_z H_z(k)} \{N_\ell(k) - Y(k) \alpha_{z\ell}(k)\}.$$

Differentiating $F_{zj}(\tau) = \sum_{k=1}^L S_z(t_k -) \alpha_{zj}(k)$ gives

$$dF_{zj}(\tau) = \sum_{k=1}^L S_z(t_k -) d\alpha_{zj}(k) - \sum_{k=1}^L \frac{F_{zj}(\tau) - F_{zj}(t_k)}{1 - \alpha_{z\cdot}(k)} d\alpha_{z\cdot}(k).$$

Substitution yields (1). The treatment-proportion derivative is $I(Z = z)(A - p_z)/\pi_z$. Since h_b is linear in θ_P , its influence function is $g_{b,P} = \Phi_{N,P} - b\Phi_{\kappa,P}$, proving the stated uniform asymptotic-linear expansion.

The plug-in influence functions are continuous functions of $\hat{q}$. Uniform consistency of $\hat{q}$, boundedness, and the finite-dimensional law of large numbers imply

$$\sup_{b \in [-1,1]} |\mathbb{P}_n \hat{g}_b^2 - E_P g_{b,P}^2| = o_P(1)$$

uniformly over $\mathcal{P}$. At $b = \psi(P)$, identification implies $h_b(P) = N_j - \psi(P)\kappa = 0$. The influence functions are uniformly bounded under A2 and A7, and their variances at the truth are bounded below under A8. Hence the Berry–Esseen bound applies uniformly:

$$\sup_{P \in \mathcal{P}} \sup_x \left| \Pr_P \left[\frac{\sum_{i=1}^n g_{\psi(P),P}(O_i)}{\sqrt{n E_P g_{\psi(P),P}^2}} \leq x \right] - \Phi(x) \right| \rightarrow 0.$$

Combining this result with asymptotic linearity and variance consistency yields $T_n\{\psi(P)\} \Rightarrow \chi_1^2$ uniformly over $\mathcal{P}$. Inverting the test proves coverage. No step divides by $\kappa(P)$.

B Reproducibility

Run

```
python simulate_weak_iv_cif_extended.py --output extended_results_final \
--replications 2000 --primary-replications 5000 --workers 4
python make_simulation_figures.py
```

The simulation writes aggregate results, replicate-level diagnostics, and JSON metadata including software versions, arguments, seed, quadrature stability, and wall-clock runtime. Replicate output contains both score components, quadratic coefficients, variance terms, $\hat{N}$, $\hat{\kappa}$, and Wald endpoints. The figure script reconstructs every plotted quantity from those files. No training or tuning data are used because no machine-learning nuisance estimator is fitted.